

Engineering Statistics

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Chapter 11: Linear Models

Outline

- ▶ Motivation for linear models.
- ▶ Examples of linear regression.
- ▶ Model formulation and parameter estimation.
- ▶ Inference for linear regression.
- ▶ Introduction to ANOVA.
- ▶ ANOVA formulation and the F test.

Motivation for Linear Regression

In many scientific and engineering problems, we want to understand how a response variable changes with one or more explanatory variables.

Examples:

- ▶ How does the **construction cost** depend on the **building size**?
- ▶ How does the **project duration** depend on the **number of workers**?
- ▶ How does the **material strength** depend on the **curing temperature**?

Main Objectives

- ▶ **Understand relationships** between variables.
- ▶ **Quantify effects**: how much does the response change when a predictor changes?
- ▶ **Prediction**: estimate the response for new observations.

Linear regression provides a simple and widely used framework to model these relationships.

Motivation for ANOVA

In many experiments, we want to compare the means of several groups to determine whether they differ significantly.

Examples:

- ▶ Do different **construction materials** produce different average strengths?
- ▶ Do different **curing temperatures** affect the average strength of concrete?
- ▶ Do different **construction methods** lead to different project completion times?

A naive approach is to conduct multiple pairwise t -tests, but this leads to an increased probability of false positives.

Analysis of Variance (ANOVA)

- ▶ **Analysis of Variance (ANOVA)** provides a systematic framework to test

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$$

against the alternative that at least one group mean differs.

- ▶ The key idea is to compare **variation between groups** with **variation within groups** using the F statistic.
- ▶ Linear regression and ANOVA are both special cases of the general linear model, with linear regression modeling relationships with continuous predictors and ANOVA comparing group means using categorical predictors.

Example: Simple Linear Regression

- ▶ We are interested in studying the relationship between the sales y of a product and the spending x on advertisements on YouTube.
- ▶ We observe n pairs of data $\{y_i, x_i\}, i = 1, \dots, n$, where y_i denotes the sales and x_i denotes the advertising spending for observation i .
- ▶ The simple linear regression model is

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad i = 1, \dots, n. \quad (1)$$

- ▶ y_i : response variable (sales)
 - ▶ x_i : predictor variable (advertising spending)
 - ▶ ϵ_i : random error (unexplained variation)
- ▶ The model represents a straight line with intercept β_0 and slope β_1 .

Least Squares Illustration

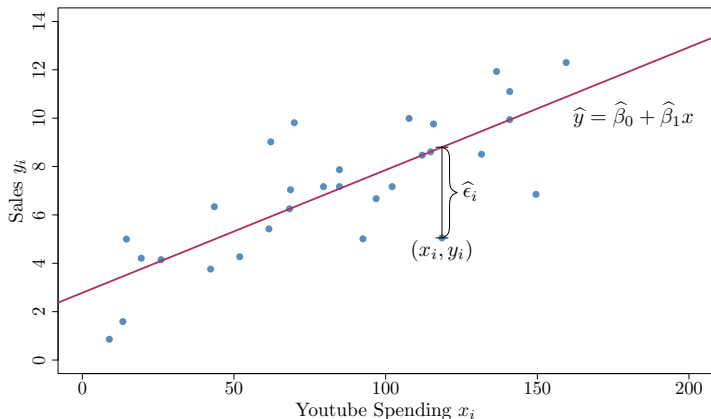


Figure: Illustration of the least-squares method based on the sales data. The regression line is $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$. The i th data point is represented by (x_i, y_i) and its vertical distance from the regression line is $\hat{\epsilon}_i$.

Least Squares Estimation

- ▶ A common approach is the **least squares (LS)** method.
- ▶ The LS method chooses β_0 and β_1 to minimize the sum of squared vertical distances from the data points to the fitted line.
- ▶ Algebraically, LS minimizes

$$\ell(\beta_0, \beta_1) = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2.$$

- ▶ Using calculus, we obtain the LS estimates $\hat{\beta}_0$ and $\hat{\beta}_1$.
- ▶ The fitted regression line is

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x.$$

Formulas

- ▶ The LS estimator of the slope is

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}.$$

- ▶ The LS estimator of the intercept is

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Example: Multiple Linear Regression

- ▶ In addition to YouTube advertising spending (x_{i1}), we may also have data on advertising spending on Facebook (x_{i2}) and in newspapers (x_{i3}).
- ▶ The dataset contains observations

$$\{y_i, x_{i1}, x_{i2}, x_{i3}\}, \quad i = 1, \dots, n,$$

where y_i denotes the product sales for observation i .

- ▶ To study how these advertising channels jointly affect sales, we consider the multiple linear regression model

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \epsilon_i, \quad i = 1, \dots, n.$$

- ▶ This model allows us to assess the effect of each advertising channel on sales.

Scatterplot of Marketing Data

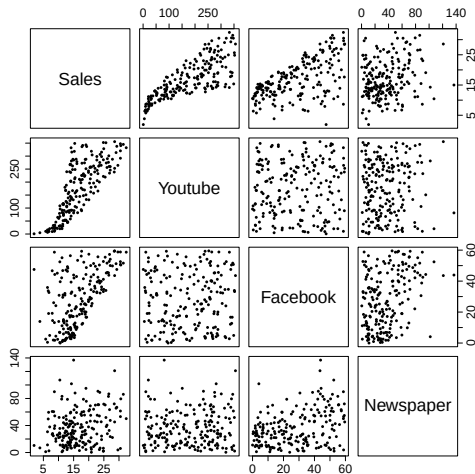


Figure: Scatterplots for pairs of variables in the marketing data.

Least Squares for Multiple Linear Regression

- ▶ To fit the multiple linear regression model, we again use the **least squares (LS)** method.
- ▶ LS chooses the parameter values that minimize the sum of squared vertical distances from the data points to the regression hyperplane

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3.$$

- ▶ Algebraically, LS minimizes

$$\ell(\beta_0, \beta_1, \beta_2, \beta_3) = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \beta_2 x_{i2} - \beta_3 x_{i3})^2.$$

Confidence Intervals for Regression Coefficients

- ▶ Consider the simple linear regression model, a confidence interval for the slope β_1 is

$$\hat{\beta}_1 \pm t_{n-2, 1-\alpha/2} \sqrt{\frac{\hat{\sigma}^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}.$$

- ▶ A confidence interval for the intercept β_0 is

$$\hat{\beta}_0 \pm t_{n-2, 1-\alpha/2} \sqrt{\hat{\sigma}^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right)}.$$

- ▶ Here $\hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^n (y_i - \hat{y}_i)^2$ and $t_{n-2, 1-\alpha/2}$ is the t critical value.
- ▶ For multiple regression, formulas are more complicated. We rely on software to do that.

Parameter Estimates for Marketing Data

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	3.5266	0.3742	9.422	<2e-16	***
youtube	0.0457	0.0013	32.809	<2e-16	***
facebook	0.1885	0.0086	21.893	<2e-16	***
newspaper	-0.0010	0.0058	-0.177	0.86	

	2.5%	97.5%
(Intercept)	2.789	4.265
youtube	0.043	0.049
facebook	0.172	0.206
newspaper	-0.013	0.011

Prediction in Simple Linear Regression

- ▶ Consider the simple linear regression model, after estimating $\hat{\beta}_0$ and $\hat{\beta}_1$, the fitted regression line is

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x.$$

- ▶ To predict the response at a new value x_0 , we use

$$\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0.$$

- ▶ Interpretation: $\hat{y}_0$ is the predicted mean response when the predictor takes value x_0 .
- ▶ Example: if x represents advertising spending and y represents sales, then $\hat{y}_0$ gives the predicted sales for spending level x_0 .

Prediction for Market Data

New covariate value:

youtube	facebook	newspaper
176.45	27.92	36.66

The 95% CI for the average response at this $\mathbf{x}_0$ is

Confidence interval:

estimate	lower	upper
16.83	16.55	17.11

The 95% prediction interval for a new response at this $\mathbf{x}_0$ is

Prediction interval:

estimate	lower	upper
16.83	12.83	20.83

The prediction interval is, of course, wider.

Model Diagnostics for Linear Regression

- ▶ After fitting a linear regression model, we should check whether the model assumptions are reasonable.
- ▶ Key assumptions:
 - ▶ Linearity: the mean of y is a linear function of the predictors.
 - ▶ Constant variance (homoscedasticity): $\text{Var}(\epsilon_i) = \sigma^2$.
 - ▶ Normality of errors.
 - ▶ Independence of observations.

Diagnostic plots

- ▶ Diagnostic plots help assess these assumptions:
 - ▶ Residuals vs fitted values (check linearity and constant variance).
 - ▶ Normal Q–Q plot (check normality of errors).
 - ▶ Scale–location plot (check variance stability).
 - ▶ Residuals vs leverage (identify influential observations).
- ▶ If assumptions are violated, model refinement or transformation may be needed.

Introduction to ANOVA

- ▶ In many experiments, we want to determine whether several groups have the same mean.
- ▶ Example: Do different doses of Vitamin C lead to different tooth growth?
- ▶ ANOVA provides a statistical method to compare the means of multiple groups.
- ▶ Consider the one-way ANOVA model

$$y_{ij} = \mu + \alpha_i + \epsilon_{ij}, \quad i = 1, \dots, k, \quad j = 1, \dots, n_i.$$

- ▶ The main hypothesis tested in ANOVA is

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_k,$$

which states that all group means are equal.

Example: One-Way ANOVA (Tooth Growth)

- ▶ A study investigates the effect of different doses of Vitamin C on tooth growth in guinea pigs.
- ▶ Three treatment levels are used: 0.5, 1, and 2 (units of Vitamin C). A total of $3n$ guinea pigs are used, with n pigs assigned to each treatment group.
- ▶ Let y_{ij} denote the tooth growth length for the j th pig receiving treatment i , where

$$i = 1, 2, 3, \quad j = 1, \dots, n.$$

- ▶ This can be modeled using a **one-way ANOVA model**

$$y_{ij} = \mu + \alpha_i + \epsilon_{ij}.$$

- ▶ Here: μ is the overall mean tooth growth, α_i is the effect of treatment i , ϵ_{ij} is the random error term.

Visualization of Data

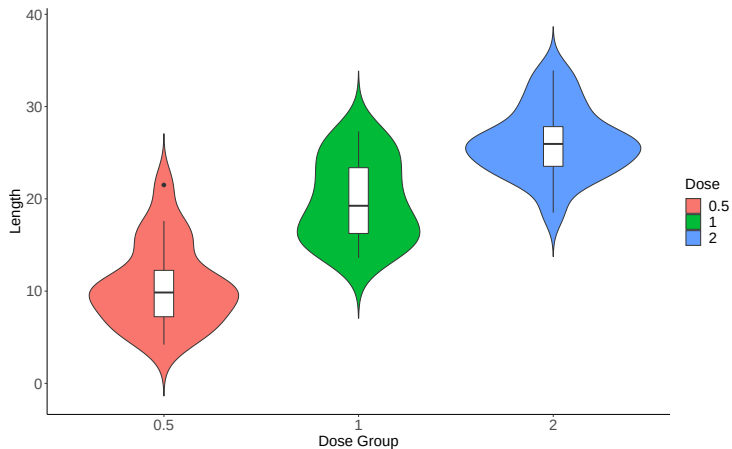


Figure: Violin plots that visualize the distributions of the three dose groups in the tooth growth data.

F Test in One-Way ANOVA

- ▶ The key idea is to compare the variation between groups with the variation within groups using an F test.
- ▶ Decompose the total variation:

$$SST = SSB + SSE,$$

where SSB = variation between groups and SSE = variation within groups.

- ▶ Compute mean squares

$$MSB = \frac{SSB}{k-1}, \quad MSE = \frac{SSE}{n-k}.$$

- ▶ Test statistic:

$$F = \frac{MSB}{MSE}.$$

- ▶ Under H_0 , $F \sim F_{k-1, n-k}$.

One-Way ANOVA Table

- ▶ These quantities are summarized in the ANOVA table.

Source	DF	Sum of Squares	Mean Square	F
Between Groups	$k - 1$	SSB	$MSB = \frac{SSB}{k - 1}$	$\frac{MSB}{MSE}$
Within Groups	$n - k$	SSE	$MSE = \frac{SSE}{n - k}$	
Total	$n - 1$	SST		

Data Analysis

```
> dat=read.csv(file="ToothGrowth.csv", header=T)

# Fit one-way ANOVA
> model <- aov(len ~ factor(dose), data = dat)

> summary(model)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
factor(dose)	2	2426	1213	67.42	9.53e-16 ***
Residuals	57	1026	18		

```
---
```

Multiple Comparisons

- ▶ After a significant ANOVA result, we often want to know *which* group means differ.
- ▶ Tukey's Honest Significant Difference (HSD) provides a method for making all pairwise comparisons between group means.
- ▶ It controls the overall Type I error rate when performing multiple comparisons.
- ▶ For groups i and j , we compare

$$|\bar{y}_i - \bar{y}_j|$$

to the Tukey critical value.

- ▶ If the difference exceeds the critical value, the two group means are significantly different.

Data Analysis

```
> TukeyHSD(model)
  Tukey multiple comparisons of means
    95% family-wise confidence level
```

```
Fit: aov(formula = len ~ factor(dose), data = dat)
```

```
$`factor(dose)`
```

	diff	lwr	upr	p adj
1-0.5	9.130	5.901805	12.358195	0.00e+00
2-0.5	15.495	12.266805	18.723195	0.00e+00
2-1	6.365	3.136805	9.593195	4.25e-05

Example: Two-Way ANOVA (Job Satisfaction)

- ▶ A study investigates whether job satisfaction is related to education level and gender.
- ▶ Data are collected from $n = 58$ individuals.
- ▶ Education has three levels: no high school degree, high school/some college, and college degree. Gender has two levels: Male and Female.
- ▶ Let y_{ijk} denote the satisfaction score for the k th individual with education level i and gender j .
- ▶ A two-way ANOVA model is

$$y_{ijk} = \mu + \alpha_i + \beta_j + \epsilon_{ijk},$$

where α_i is the education effect and β_j is the gender effect.

Visualization of Data

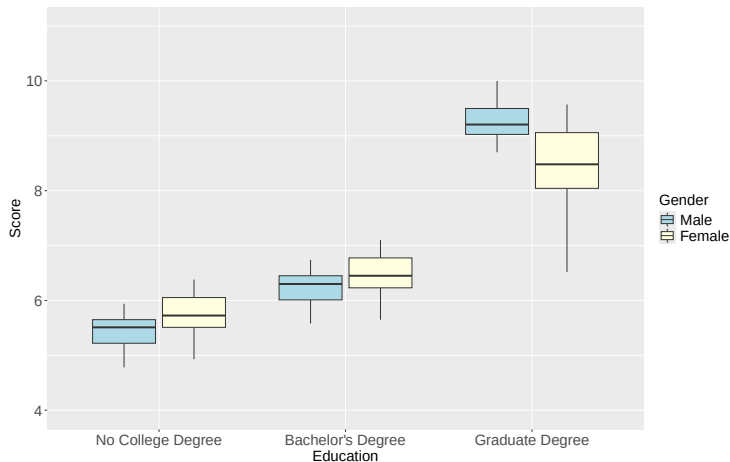


Figure: Boxplots of the job satisfaction data grouped by education and gender level combinations.

Two Way ANOVA model

- ▶ Model with interaction

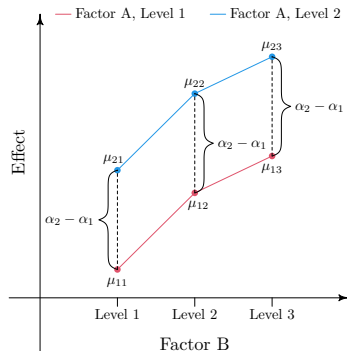
$$y_{iju} = \mu + \alpha_i + \beta_j + \gamma_{ij} + \epsilon_{iju}, \quad (2)$$

for $i = 1, \dots, a; j = 1, \dots, b; u = 1, \dots, n_{ij}$.

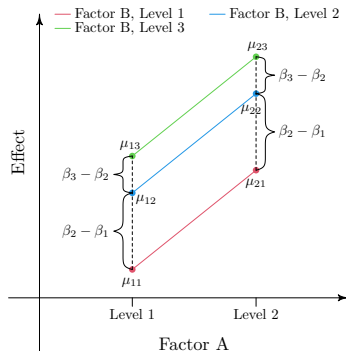
- ▶ Model without interaction

$$y_{iju} = \mu + \alpha_i + \beta_j + \epsilon_{iju} \quad (3)$$

Without Interaction



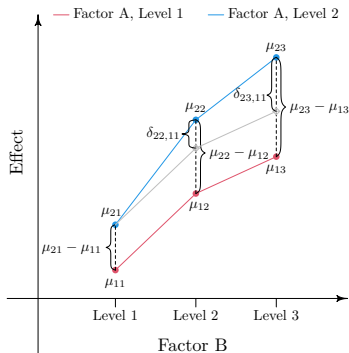
(a) Effect of A given B



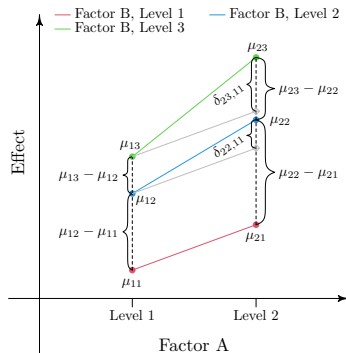
(b) Effects of B given A

Figure: Illustration of (a) the effect of factor A at each level of factor B, and (b) the effects of factor B at each level of factor A, in a 2×3 ANOVA main effects model. The effect of changing the level of one factor is the same at each level of the other factor.

With Interaction



(a) Effect of A given B



(b) Effects of B given A

Figure: Illustration of (a) the effect of factor A at each level of factor B, and (b) the effects of factor B at each level of factor A, in a 2×3 ANOVA model with interaction present.

Two-Way ANOVA Table (with Interaction)

- ▶ In a two-way ANOVA with interaction, the total variation is decomposed into
 - ▶ variation due to factor A,
 - ▶ variation due to factor B,
 - ▶ variation due to the interaction between A and B,
 - ▶ residual (error) variation.

Source	DF	Sum of Squares	Mean Square
Factor A	$a - 1$	SSA	$MSA = \frac{SSA}{a - 1}$
Factor B	$b - 1$	SSB	$MSB = \frac{SSB}{b - 1}$
Interaction (A×B)	$(a - 1)(b - 1)$	$SSAB$	$MSAB = \frac{SSAB}{(a - 1)(b - 1)}$
Error	$N - ab$	SSE	$MSE = \frac{SSE}{N - ab}$
Total	$N - 1$	SST	

Data Analysis

```
> job.model <- aov(score ~ factor(gender)*factor(education_level), data = job.dat)
> summary(job.model)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
factor(gender)	1	0.54	0.54	1.787	0.18709
factor(education_level)	2	113.68	56.84	187.892	< 2e-16 ***
factor(gender):factor(education_level)	2	4.44	2.22	7.338	0.00156 **
Residuals	52	15.73	0.30		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Two-Way ANOVA Table (No Interaction)

- If interaction is not significant.

Source	DF	Sum of Squares	Mean Square	F
Factor A	$a - 1$	SSA	$MSA = \frac{SSA}{a - 1}$	$\frac{MSA}{MSE}$
Factor B	$b - 1$	SSB	$MSB = \frac{SSB}{b - 1}$	$\frac{MSB}{MSE}$
Error	$N - a - b + 1$	SSE	$MSE = \frac{SSE}{N - a - b + 1}$	
Total	$N - 1$	SST		

Data Analysis

► Assuming no interactions.

```
> job.model <- aov(score ~ factor(gender)+factor(education_level), data = job.dat)
> summary(job.model)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
factor(gender)	1	0.54	0.54	1.447	0.234
factor(education_level)	2	113.68	56.84	152.172	<2e-16 ***
Residuals	54	20.17	0.37		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1